

# v1.0Boson Deep Dive Architecture Emulation

## V1.0 to V2.0 Versioning Delta — Tardigradia SubParticles Engine

**Architecture:** One-Boson Universe Architecture **Type:** Scientific Change Log | **Date:** June 2026  
**Applies to:** v1.0Boson Deep Dive Architecture Emulation V2.0.md

### Revision Necessity — Four Drivers

1. **Experimental discoveries 2022-2026** — LHC Run 3, LIGO O4, KATRIN 2024, NIF ignition, and other landmark results supersede V1.0 physics parameters
2. **New confirmed particles and phenomena** — Pines’ Demon (2023), HH production evidence (2024), IceCube tau neutrino (2023), and others add entirely new simulation modes
3. **Platform upgrade** — WebGPU (Chrome 113+ 2023) replaces WebGL; enables  $10^{17+}$  particles
4. **Precision improvements** — CODATA 2022 constants, FLAG 2024 lattice QCD, NuFIT 5.3 oscillations

### V1.0 vs V2.0 Specific Changes

| Parameter                          | V1.0                      | V2.0                                                                                                          | Severity | Consequence if Not Updated                                         |
|------------------------------------|---------------------------|---------------------------------------------------------------------------------------------------------------|----------|--------------------------------------------------------------------|
| <b>BEC temperature record</b>      | General BEC description   | MIT 2024: 500 pK sodium BEC — coldest matter record. $kT < \hbar\omega_{\text{trap}}$ threshold implemented   | LOW      | Updates extreme limit parameter for BEC simulation mode            |
| <b>Photon BEC</b>                  | Photons noted as massless | Bonn 2023: room-temperature photon BEC — photons acquire effective mass via cavity confinement                | HIGH     | Entirely new particle mode: PHOTON_MASSIVE_BEC not present in V1.0 |
| <b>Boson occupancy rendering</b>   | Standard instancing       | V2.0: WGSL atomicAdd for cell occupancy — N bosons in same cell renders as $N \times$ brightness accumulation | MEDIUM   | V1.0 rendering cannot show bosonic pile-up behavior correctly      |
| <b>Quantum vortex reconnection</b> | Superfluidity mentioned   | LMSS 2024: reconnection rate $\Gamma = \kappa^2$ — adds VORTEX particle type                                  | MEDIUM   | Missing quantum vortex dynamics in superfluid boson simulation     |

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## Simulation Impact Summary

The changes above drive the following modifications to the game engine architecture:

- **New particle types:** Photon
  - **Parameter updates:** All values in `static constexpr` declarations refreshed from 2022-2026 data
  - **WebGPU migration:** Particle update loop moves from CPU JavaScript to WGSL compute shaders
  - **New interaction modes:** Triggered by new experimental discoveries
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## Unchanged Architecture

All foundational philosophy is **carried forward unchanged:** - Worldline Monism (One-Particle Universe) ontology - Proper time  $\tau$  as primary particle identifier - Intrinsic vs. Extrinsic property distinction - Benevolence metric ( $C \cdot F \cdot S$  product formula) - Object-Oriented data structure (struct ParticleInstance extends to V2.0)

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*End of Versioning Delta | v1.0Boson Deep Dive Architecture Emulation V1.0 archived | V2.0 is the active specification as of June 2026*